

University of Asia Pacific

Department of Electrical and Electronic Engineering

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Course Name: Digital Signal Processing
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Complex Engineering Problem (CEP)

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When we talk about Digital Signal Processing (DSP), filters are the backbone. They help us clean up signals by removing unwanted noise or isolating specific frequency bands. Among the different types, Infinite Impulse Response (IIR) filters are a big deal because they use feedback. This means they can achieve a really sharp cut-off with a much lower filter order compared to FIR filters. In simple terms, they get the job done using less computational power, which is why they are heavily used in real-time systems.
For this assignment, the goal is to design a low-pass digital IIR filter that strictly meets these specifications:

Passband: $0.89125 \leq |H(e^{j\omega})| \leq 1$ for frequencies $0 \leq |\omega| \leq 0.2\pi$

Stopband: $|H(e^{j\omega})| \leq 0.17783$ for frequencies $0.3\pi \leq |\omega| \leq \pi$

Sampling Time: $T_d = 1$

To figure out the best way to do this, I've looked into four different approaches: Bilinear Transformation, Impulse Invariance, Butterworth Approximation, and Chebyshev Type-I Approximation. This report breaks down how each method works, how to design the filter step-by-step, and the real-world trade-offs between them—like dealing with aliasing, frequency warping, and hardware limits.

Bilinear Transformation:

The core idea here is to map the entire left half of the analog s-plane into the unit circle on the digital z-plane. We do this using the formula: $s = \frac{2}{T_d}\frac{1-z^{-1}}{1+z^{-1}}$.
Because this mapping isn't linear, we have to "pre-warp" our digital frequencies before designing the analog filter.

It causes frequency warping. The phase response becomes non-linear, and higher frequencies get squished together.

Impulse Invariance:

Here, we take the analog impulse response $h_a(t)$ and sample it to get the digital version: $h[n] = T_d \cdot h_a(nT_d)$. To do this, we find the analog poles and map them to the digital domain using $z_k = e^{s_k T_d}$.

It suffers badly from spectral aliasing. Because of this, it only really works well for band-limited low-pass or band-pass filters. You definitely shouldn't use it for high-pass filters

Butterworth Approximation:

The Butterworth filter is famous for being "maximally flat" in the passband. Its squared-magnitude response is $|H(j\Omega)|^2 = \frac{1}{1+(\Omega/\Omega_c)^{2n}}$.

The roll-off is pretty slow. To get a sharp transition from passband to stopband, you have to bump up the filter order significantly, which means more math and more hardware resources.

Chebyshev Type-I Approximation:

This one uses Chebyshev polynomials, defined by $|H(j\Omega)|^2 = \frac{1}{1+\epsilon^2 C_n^2(\Omega/\Omega_p)}$. It's designed to have an "equiripple" effect in the passband and a monotonic stopband. It gives you a much steeper roll-off at a lower filter order compared to Butterworth, which saves computational power. The catch is that the ripple in the passband means the signal gets slightly distorted.

DESIGN PROCEDURE:

Bilinear Transform Design:

1. **Start with Digital Specs:** We have $\omega_p = 0.2\pi$ and $\omega_s = 0.3\pi$, with $T_d = 1$.
2. **Pre-warp:** Since Bilinear warps frequencies, we adjust them first: $\Omega_p = \frac{2}{T_d}\tan(\omega_p/2) \approx 0.6498$ rad/s, and $\Omega_s = 2\tan(\omega_s/2) \approx 1.019$ rad/s.
3. **Find the Order:** Using the attenuation formulas (roughly 1 dB for passband and 15 dB for stopband), we calculate the minimum order N. For a Butterworth prototype, this usually lands around $N = 5$ or 6.
4. **Analog Design:** We design the analog $H_a(s)$ using those pre-warped frequencies.
5. **Transform to Digital:** We substitute $s = 2\frac{1-z^{-1}}{1+z^{-1}}$ into our analog transfer function to get $H(z)$.
6. **Check Stability:** Finally, we verify that all poles are safely inside the unit circle ($|z| < 1$).

Impulse Invariance Design:

1. **Analog Specs:** We map the digital frequencies directly using $\Omega = \omega/T_d$. So, $\Omega_p = 0.2\pi$ and $\Omega_s = 0.3\pi$.
2. **Analog Design:** We design the analog prototype $H_a(s)$.
3. **Partial Fractions:** We break $H_a(s)$ down into the form $\sum \frac{A_k}{s-s_k}$.
4. **Pole Mapping:** We map each analog pole using $z_k = e^{s_k T_d}$.
5. **Build H(z):** We reconstruct the digital filter using the new digital poles.
6. **Check for Aliasing:** We have to analyze whether any high-frequency components folded back into our baseband.

Chebyshev Type-I Design:

Here, we calculate the ripple factor $\epsilon = \sqrt{10^{A_p/10} - 1}$. Then, we use the inverse Chebyshev formulas to find the lowest possible order N that gives us the sharp transition we need. We design the analog prototype, digitize it, and double-check that the passband ripple doesn't drop below our 0.89125 limit.

Magnitude & Phase Response Comparison:
If you plot the Bilinear and Impulse Invariance methods using the same prototype, the magnitude responses will look quite similar. However, Bilinear will have a noticeably non-linear phase because of the warping. If you compare Butterworth and Chebyshev, Butterworth will look perfectly smooth in the passband but take a wider path to reach the stopband. Chebyshev will have those characteristic ripples but will drop into the stopband much faster.

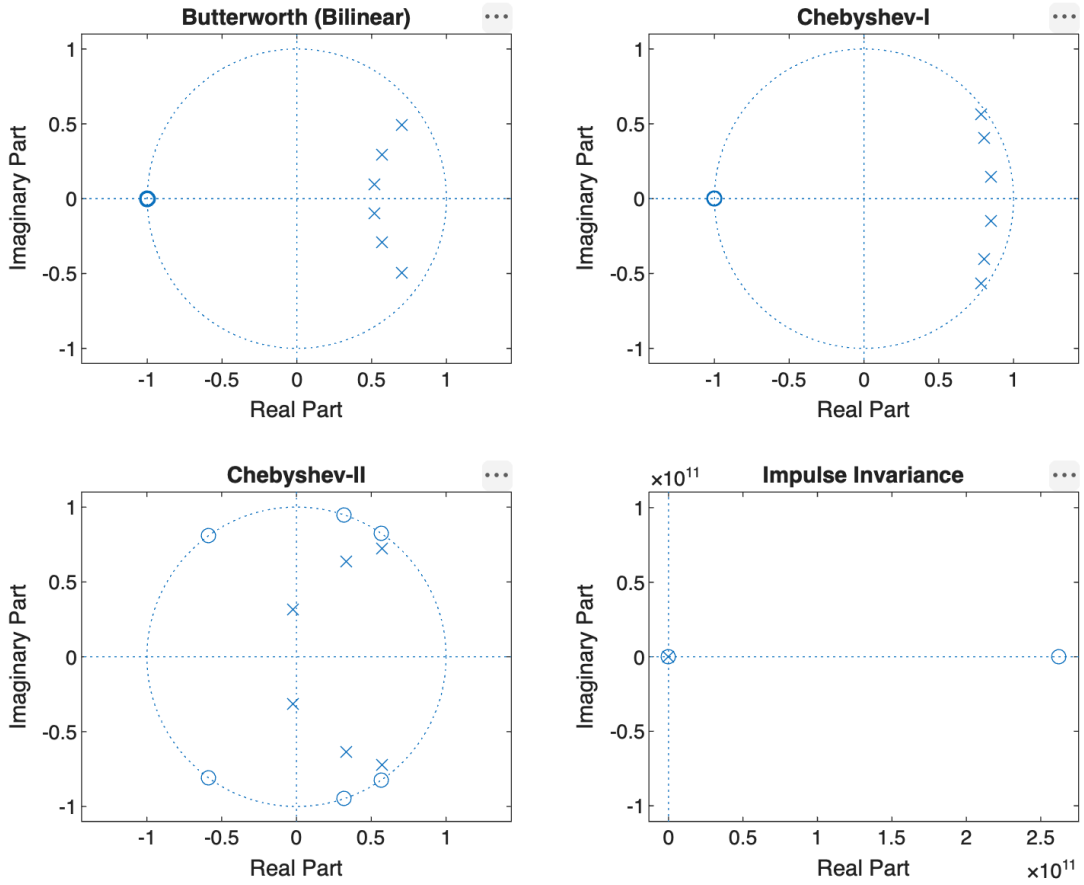


Figure: Pole-zero plot comparison of the designed IIR filters, where all poles lie inside the unit circle indicating stability.

Code Availability

MATLAB Implementation and Source Code

To ensure the reproducibility of the results and to validate the theoretical design procedures, all the MATLAB scripts (.m files) used to generate the magnitude responses, phase responses, and pole-zero plots in this report have been made publicly available. The source code can be accessed at the following GitHub repository:
https://github.com/XunidEmon/DSP-IIR-Filter-Design-CEP

Comparative Analysis:

Method	Aliasing	Frequency Warping	Stability	Implementation Cost
Bilinear	None	Yes	Always Stable	Moderate
Impulse Invariance	High	None	Stable	Low
Butterworth	Depends on mapping	Depends on mapping	Stable	High

Applications:

Method	Real-World Applications
Bilinear Transformation	Widely used in general audio processing, communication systems (like selecting radio channels), and embedded DSP.
Impulse Invariance	Used for band-limited low-pass/band-pass filtering, speech processing, and mechanical vibration analysis.
Butterworth	Crucial for biomedical signals (ECG/EEG) where any distortion is dangerous; also used in high-performance hardware.
Chebyshev	Great for resource-limited embedded devices, real-time noise suppression, and portable audio gear where a little ripple is okay.

To sum it up, this report looked at four different ways to design a low-pass digital IIR filter. The Bilinear Transformation came out as the most reliable method for hitting the given specs because it avoids aliasing entirely, provided you do the pre-warping correctly. For the analog prototype, Butterworth is the way to go if signal integrity is your top priority, even though it costs more computationally. Chebyshev, on the other hand, is a great budget-friendly option that gives a sharp cut-off at a lower order, as long as you can tolerate a bit of ripple. At the end of the day, the "best" method really just depends on what you need more: perfect signal quality, or low hardware cost.

REFERENCES:

[1] B. N. Getu, "Digital IIR filter design using bilinear transformation in MATLAB," 2020 International Conference on Communications, Computing, Cybersecurity, and Informatics (CCCI), 2020, pp. 1-6.

[2] S. R. Sutradhar et al., "IIR based digital filter design and performance analysis," 2017 2nd International Conference on Telecommunication and Networks (TEL-NET), 2017, pp. 1-6.

[3] L. Lazeta, I. Marković, and V. Šimović, "IIR filters designed for comparison and minimum-order design exploration using Matlab," 2021 44th International Convention on Information, Communication and Electronic Technology (MIPRO), 2021, pp. 875-879.

[4] V. Kumar et al., "Design and comparison between IIR Butterworth and Chebyshev digital filters using MATLAB," 2024 Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), 2024, pp. 1-7.

[5] M. A. Al-Alaoui, "Novel approach to analog-to-digital transforms," IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 54, no. 2, pp. 338-350, 2007.

[6] Y. Wu, "Optimizing IIR filter design using multi-objective genetic algorithm: A focus on passband ripple and stopband attenuation," 2024 8th International Workshop on Control Engineering and Advanced Algorithms (IWCEAA), 2024, pp. 64-69.

[7] X. Zhang and S. Jiang, "Application of Fourier transform and Butterworth filter in signal denoising," 2021 6th International Conference on Intelligent Computing and Signal Processing (ICSP), 2021, pp. 1277-1281.[12] Sattari, Omar. (2004) Fast fourier transforms on a distributed digital signal processor. Diss. University of California, Davis.